



Letter of Support – SterileZone Guardian

05/15/2026

To Whom It May Concern,

I am pleased to provide this letter in support of the student team competing in the HOSA Medical Innovation International event with their project focused on the development of an artificial intelligence (AI)-based Sterile Field Monitor.

As a medical device manufacturer, I see expanded value beyond the initial vision to monitor the sterile boundary. Loss of sterility to packaged devices can introduce infections that are more likely to lead to sepsis. Current inspection practices rely heavily on manual visual inspection by operating room personnel, which may be subject to variability due to time constraints, environmental conditions, and human factors. Development of automated optical inspection technologies is an area of growing interest within the medical device industry, particularly for applications involving defect detection, seal inspection, and packaging quality evaluation. Exploring these technologies in the context of point-of-use inspection within the operating room is highly relevant.

The proposed application of AI-assisted optical monitoring has the potential to improve consistency, support clinical decision-making, and enhance patient safety by providing an additional layer of quality assessment not only for monitoring the sterile boundary but also when applied immediately before a product is introduced into the sterile field

What makes this project particularly impressive is the team's ability to combine concepts from healthcare, biomedical engineering, artificial intelligence, and quality assurance into a practical and clinically relevant application. Their work reflects a strong understanding of both the technical and real-world considerations associated with the operating room.

In addition to the technical aspects of the project, the students have demonstrated initiative, creativity, professionalism, and a genuine interest in solving meaningful healthcare problems. The project exemplifies the type of interdisciplinary thinking and innovation encouraged through HOSA's Medical Innovation program.

I strongly support this team and believe their project represents a thoughtful and promising contribution to healthcare technology innovation. I commend their dedication, curiosity, and commitment to improving patient care through engineering and scientific problem solving.



Please feel free to contact me if additional information is required.

Regards,

A handwritten signature in black ink that reads 'Holle L. Carey'. The signature is written in a cursive style with a large, stylized 'H' and a prominent 'C'.

Holle Carey
Sr. Regulatory Engineer
University of Texas at Dallas -Texas Biomedical Device Center